

Erratum: Source depth estimation based on synthetic aperture beamforming for a moving source [J. Acoust. Soc. Am. 138, 1678–1686 (2015)]

[T. C. Yang](#)

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Erratum: Source depth estimation based on synthetic aperture beamforming for a moving source [J. Acoust. Soc. Am. 138, 1678–1686 (2015)]

T. C. Yang^{a)}

College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, 310058, China

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Page: 3075

While it is not obvious, it was realized after the publication of this paper that Eq. (1), the expression of synthetic array beamforming, has implicitly assumed that the range integration is synchronized with the range of the data, as is the case for the Hankel transform. In real data, as pointed out in Ref. 1, the range of the data and the range of synthetic beamforming can differ by an “offset-range,” $\delta = r_1 - r_1^{data} = r_2 - r_2^{data}$, where r_1 and r_2 are the ranges of synthetic beamforming or limits of the range integration, and r_1^{data} , r_2^{data} are the (true) source ranges at the beginning and end of the data. In this case, the spectral peak at $k_r = k_m$ is given by $g(k_m, z_r) = b_m \phi_m(z_r) e^{ik_m \delta}$ in contrast with Eq. (5) given in the paper. The phase $e^{ik_m \delta}$ has no effect on the

wavenumber spectral density but does affect the depth ambiguity function. The simulations and data analysis presented in the paper had assumed the use of Eq. (5), or that δ is zero. The statement in the conclusions that “The advantage of the proposed method, Eqs. (6) or (9), is that the source range is not known nor is it a required input” is not strictly correct. (This is the author’s error.) In processing real data, the source initial range, or range offset δ , needs to be searched for as in Ref. 1.

¹T. C. Yang and W. Xu, “Data-based depth estimation of an incoming autonomous underwater vehicle,” *J. Acoust. Soc. Am.* **140**, EL302–EL306 (2016).

^{a)}Electronic mail: tsihyang@gmail.com